Cardiac Cycle Overview

The cardiac cycle is the sequence of electrical and mechanical events during one heartbeat.

Systole is the phase of ventricular contraction that ejects blood into the aorta and pulmonary artery.

Diastole is the phase of ventricular relaxation during which the ventricles fill with blood.

Cardiac output equals stroke volume multiplied by heart rate, for example 70 mL x 72 beats per minute = 5040 mL per minute.

Phases of the Ventricular Cycle

The ventricular cycle proceeds through four phases in a fixed order:

1. Isovolumetric contraction closes the atrioventricular valves and raises ventricular pressure.

2. Ejection opens the semilunar valves and ejects blood into the arteries.

3. Isovolumetric relaxation closes the semilunar valves while pressure falls.

4. Filling opens the atrioventricular valves and the ventricles refill with blood.

Left versus Right Ventricle

The left ventricle has a thick muscular wall whereas the right ventricle has a thinner wall.

Left ventricular pressure during systole reaches about 120 mmHg, unlike the right ventricle which reaches about 25 mmHg.

The left ventricle supplies the systemic circulation while the right ventricle supplies the pulmonary circulation.

A patient presenting with exertional dyspnea and a displaced apex beat may have left ventricular hypertrophy, which is diagnosed by echocardiography.